

Mathematical Foundations for SVMs

Lagrange Multipliers and KKT Conditions

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Unconstrained vs. Constrained Optimization

- **Unconstrained Optimization:** Find $\min_x f(x)$ or $\max_x f(x)$.
- *Solution:* Compute the gradient and set it to zero, $\nabla f(x) = 0$.
- **Constrained Optimization:** What if we restrict the domain?
- *Example:* Minimize $f(x, y) = x^2 + y^2$ subject to $x + y = 1$.
- We cannot simply set $\nabla f(x, y) = 0$, because the global minimum $(0, 0)$ does not satisfy our constraint! We need a new mathematical tool.

Prerequisite: Understanding Contour Lines

- Before we solve constrained optimization, we need to visualize 3D functions in 2D.
- Imagine a 3D surface (like a mountain) represented by $f(x, y)$.
- A **contour line** (or level curve) connects all points where the function has the exact same value.
- Mathematically: $f(x, y) = k$ for some constant k .
- *Topographical Map Analogy*: If you walk along a contour line, your elevation never changes. If you cross contour lines, you are going uphill or downhill.

Visualization

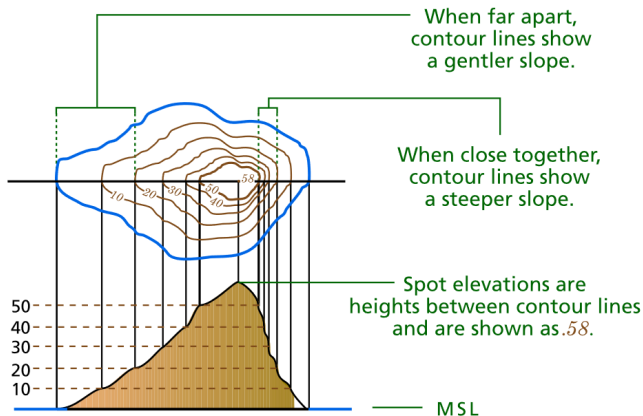


Figure: Example contour lines of a function of 2 variables.

The Gradient and the Contour Line

- The gradient vector, ∇f , points in the direction of steepest ascent.
- **Crucial Geometric Fact:** The gradient is *always* perfectly perpendicular (orthogonal) to the contour lines.
- *Why?* Because the contour line is the direction of *zero* change. The steepest way up a mountain is always 90 degrees away from the flat path.
- Keep this in mind: ∇f is an arrow pointing perpendicularly away from the contour curve.

The Geometry of the Constraint

- We want to maximize $f(x, y)$ subject to a constraint $g(x, y) = c$.
- Think of $f(x, y)$ as the mountain, and $g(x, y) = c$ as a fenced walking path on that mountain. You cannot leave the path.
- As you walk along the path g , if your path **crosses** a contour line of f , it means you are changing elevation. You can keep walking to get higher.
- The maximum elevation is reached exactly when your path **grazes** the highest possible contour line without crossing it.
- At this point, the path g and the contour line of f are **tangent**.

The Intuition Behind Lagrange Multipliers

- We want to optimize $f(x, y)$ subject to an equality constraint $g(x, y) = c$.
- Imagine walking along the constraint curve $g(x, y) = c$ overlaid on the contour lines of $f(x, y)$.
- The maximum or minimum occurs exactly where the contour line of f is **tangent** to the constraint curve g .
- If they are tangent, their normal vectors (gradients) must be parallel.
- Therefore, at the optimal point:

$$\nabla f(x, y) = \lambda \nabla g(x, y)$$

- The scalar λ is called the **Lagrange Multiplier**.

The Lagrangian Function

- To formalize this, we combine our objective and constraint into a single equation called the **Lagrangian**:

$$L(x, y, \lambda) = f(x, y) - \lambda(g(x, y) - c)$$

- To find the constrained extrema, we simply find the *unconstrained* extrema of this new Lagrangian function!
- Set the partial derivatives to zero:

$$\frac{\partial L}{\partial x} = 0 \implies \frac{\partial f}{\partial x} - \lambda \frac{\partial g}{\partial x} = 0$$

$$\frac{\partial L}{\partial y} = 0 \implies \frac{\partial f}{\partial y} - \lambda \frac{\partial g}{\partial y} = 0$$

$$\frac{\partial L}{\partial \lambda} = 0 \implies -(g(x, y) - c) = 0 \quad (\text{Recovers the constraint!})$$

Example: Solving with an Equality Constraint

Problem: Minimize $f(x, y) = x^2 + y^2$ subject to $g(x, y) = x + y = 1$.

- ① Construct the Lagrangian:

$$L(x, y, \lambda) = x^2 + y^2 - \lambda(x + y - 1)$$

- ② Take partial derivatives and set to zero:

$$\frac{\partial L}{\partial x} = 2x - \lambda = 0 \implies x = \lambda/2$$

$$\frac{\partial L}{\partial y} = 2y - \lambda = 0 \implies y = \lambda/2$$

$$\frac{\partial L}{\partial \lambda} = -(x + y - 1) = 0 \implies x + y = 1$$

- ③ Substitute x and y into the constraint equation:

$$\frac{\lambda}{2} + \frac{\lambda}{2} = 1 \implies \lambda = 1$$

- ④ Result: $x = 1/2$, $y = 1/2$. The constrained minimum is 0.5.

Moving to Inequality Constraints

- So far, we solved $\min f(x)$ subject to an equality constraint $h(x) = 0$.
- What if our constraint is an **inequality**: $g(x) \leq 0$?
- We cannot use standard Lagrange multipliers because the optimal point might be strictly inside the allowed region, where the constraint shouldn't affect the gradient at all.
- We generalize the method using the **Karush-Kuhn-Tucker (KKT) conditions**.
- For $\min f(x)$ subject to $g(x) \leq 0$, the Lagrangian is:

$$L(x, \alpha) = f(x) + \alpha g(x)$$

- There are two crucial new rules for the multiplier α :
 - 1 **Dual Feasibility**: $\alpha \geq 0$
 - 2 **Complementary Slackness**: $\alpha g(x) = 0$

The Physical Intuition of KKT

- Abstract math is easier with physical analogies. Let's look at an inequality constraint as a physical boundary.
- **The Scenario:** A dog is tied to a post. It wants to reach a bone.
- **The Objective ($f(x)$):** Minimize the distance to the bone.
- **The Constraint ($g(x) \leq 0$):** The leash has a maximum length.
 - $g(x) < 0$: There is slack in the leash (Dog is strictly inside the boundary).
 - $g(x) = 0$: The leash is perfectly taut (Dog is exactly at the limit).
 - $g(x) > 0$: The leash is broken (Mathematically invalid).
- Let α be the **force** exerted by the leash on the dog's neck.

KKT Rule 1: Dual Feasibility

The Mathematical Rule

$$\alpha \geq 0$$

- **The Physical Reality:** A leash can only pull; it cannot push.
- Because our boundary is an *inequality* constraint, the barrier can only exert force in one direction: pushing you back inside the allowed region.
- If the math allowed α to be negative, it would imply the leash was magically pushing the dog away from the post towards the bone!

KKT Rule 2: Complementary Slackness

The Mathematical Rule (The "Master Switch")

$$\alpha g(x) = 0$$

- For this product to be exactly zero, one of the two variables **must** be zero.
- **Reality 1 (Inactive Constraint):** The minimum is safely inside the region.
 - There is slack ($g(x) < 0$).
 - The boundary exerts no force ($\alpha = 0$).
- **Reality 2 (Active Constraint):** The minimum is outside the region. The optimal allowed point is stuck on the edge.
 - There is no slack ($g(x) = 0$).
 - The boundary exerts positive force to hold the point back ($\alpha > 0$).

The Mathematical Consequence: Sparsity

- What does Complementary Slackness actually do to our Lagrangian?

$$L(x, \alpha) = f(x) + \alpha g(x)$$

- Suppose we have many constraints:
 $g_1(x) \leq 0, g_2(x) \leq 0, \dots, g_N(x) \leq 0$.
- The Lagrangian becomes: $L(x, \alpha) = f(x) + \sum_{i=1}^N \alpha_i g_i(x)$
- **The Magic of KKT:** For every constraint i that is *inactive* (we didn't hit the boundary), Complementary Slackness forces $\alpha_i = 0$.
- Those terms literally **multiply by zero and vanish** from the Lagrangian.
- **Conclusion:** The final optimal solution depends *only* on the constraints we actually collide with (the active constraints). All other constraints are mathematically ignored.